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Structured and Lipschitz Bandits Over Continuous and Related Action Spaces

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Background: why structure matters beyond tabular bandits

Tabular bandits treat each arm as separate, so they learn slowly when the action space is large or continuous. Structured bandits speed this up by assuming related actions have related rewards, which lets evidence from one action help estimate others. (9 sources)

A useful starting point is the limit of the classical tabular view: standard multi-armed bandits usually assume a **finite set of independent arms**, and this becomes a poor model when the action space is very large, continuous, or internally related. In such settings, treating each action independently can force the learner to sample many near-duplicate or plainly suboptimal actions and can lead to very slow learning, or even effectively linear regret when the modeling mismatch is severe. This is exactly why later work moved to **structured action spaces**, where arms are linked by geometry, smoothness, correlations, or combinatorial constraints rather than viewed as isolated options (Eldowa et al., 2026) (Cheng et al., 2022).

The core idea is simple: when structure is present, **information gained from one arm can generalize to other arms**. Agarwal et al. make this point directly: if rewards have known structure, the reward information from one action "propagates" to others, which is what makes large or infinite action spaces tractable (Agarwal et al., 2011). They also distinguish two broad modes of sharing: **global sharing**, as in linear models where one parameter controls rewards across the whole action space, and **local sharing**, as in Lipschitz models where nearby actions must have similar rewards (Agarwal et al., 2011) (Kleinberg et al., 2008). Granmo states the same contrast in more applied terms: ordinary bandit methods are hindered in continuous or large action spaces by their inability to generalize across arms, while newer approaches explicitly model interactions among actions through global parametric links, tree search over smooth objectives, or Gaussian-process smoothing (Granmo, 2016) (Atan et al., 2014) (Bubeck et al., 2010).

This perspective also explains the main split in the literature on large action spaces. As summarized by Saito et al., one line is **parametric**, such as linear and combinatorial bandits, where expected reward is assumed to follow a known functional form; the other is **nonparametric**, such as Lipschitz bandits, where the assumption is weaker but still lets the learner exploit similarity among actions (Saito et al., 2022). Both lines are responses to the same problem: with too many actions, pure tabular learning wastes data because it cannot reuse information across related choices (Saito et al., 2022) (Eldowa et al., 2026).

The payoff from structure is better sample efficiency, but there is also a tradeoff. Nonparametric structured

methods such as Lipschitz or Gaussian-process bandits are flexible, yet they can still suffer from a **curse of dimensionality**, with rates degrading as the dimension or effective complexity of the action space grows (Agarwal et al., 2011). This is one reason the field now spans a spectrum: from tabular methods with no sharing, to local metric smoothness, to global parametric models, all the way to richer unified frameworks for stochastic structured bandits (Combes et al., 2017) (Eldowa et al., 2026).

Evidence

(Eldowa et al., 2026)

Leveraging Similarities in Multi-Armed Bandits

> *"Classical bandit models typically assume a finite set of independent arms. However, many modern applications involve structured action sets in which arms are related through geometry, correlations, or combinatorial constraints. This has motivated a substantial body of work on bandits with structured action spaces. Structured bandits extend the basic multi-armed bandit framework to large or infinite arm sets by imposing known relationships on the reward function across arms, which can be exploited to improve learning performance. Prominent examples include linear bandits (Abbasi-Yadkori et al., 2011)(Agrawal et al., 2012)(Ito et al., 2020) and combinatorial bandits (Zimmert et al., 2018)(Combes et al., 2015). A unifying treatment of several such structures in the stochastic setting was proposed by (Combes et al., 2017). Another important class is given by Lipschitz or Hölder bandits, where arms lie in a metric (possibly continuous) space and regret bounds depend on the covering or packing dimension of this space (Kleinberg et al., 2008)(Slivkins, 2011)(Kleinberg et al., 2013)Podimata and Slivkins, 2021)."*

(Cheng et al., 2022)

Distributed Task Management in Fog Computing: A Socially Concave Bandit Game

> *"In the seminal MAB problem, the set of arms is finite. In many applications, that setting only serves as an imprecise model of the situation, which potentially forces the learner to pull only suboptimal arms, thereby causing a linear regret (Trovò et al., 2016). In contrast, the Continuum-Armed Bandit (CAB) defines the set of arms over a continuous space. Such a setting has attracted intensive attention in the past few years due to its ability to model general situations. In particular, it fits the problems where the expected reward is a Lipschitz function of the arm, known as Lipschitz Bandits."*

(Agarwal et al., 2011)

Stochastic Convex Optimization with Bandit Feedback

> *"research has focused on the development of optimal and efficient algorithms for multi-armed bandits with large or even infinite action spaces, relying on various assumptions on the structure of costs (rewards) over the action space. When such a structure is present, the information about the cost of one arm propagates to other arms as well, making the problem tractable. For instance, the mean cost function is assumed to be linear in the paper (Dani et al., 2008), facilitating global "sharing of information" over a compact convex set of actions in a d-dimensional space. A Lipschitz condition on the mean cost function allows a local propagation of information about the arms, as costs cannot change rapidly in a neighborhood of an action. This has been exploited in a number of works, notably [2,(Kleinberg, 2004)(Kleinberg et al., 2008). Instead of the Lipschitz condition, Srinivas et al. [18] exploit the structure of Gaussian processes, focusing on the notion of the effective dimension. These various "non-parametric" bandit problems typically suffer from the curse of dimensionality, that is, the best possible convergence rates (after T queries) are typically of the form T α , with the exponent α approaching 1 for large dimension d."*

(Kleinberg et al., 2008)

Multi-armed bandits in metric spaces

> *"In a multi-armed bandit problem, an online algorithm chooses from a set of strategies in a sequence of n trials so as to maximize the total payoff of the chosen strategies. While the performance of bandit algorithms with a small finite strategy set is quite well understood, bandit problems with large strategy sets are still a topic of very active investigation, motivated by practical applications such as online auctions and web advertisement. The goal of such research is to identify broad and natural classes of strategy sets and payoff functions which enable the design of efficient solutions. In this work we study a very general setting for the multi-armed bandit problem in which the strategies form a metric space, and the payoff function satisfies a Lipschitz condition with respect to the metric. We refer to this problem as the "Lipschitz MAB problem". We present a complete solution for the multi-armed problem in this setting. That is, for every metric space (L,X) we define an isometry invariant Max Min COV(X) which bounds from below the performance of Lipschitz MAB algorithms for X, and we present an algorithm which comes arbitrarily close to meeting this bound. Furthermore, our technique gives even better results for benign payoff functions."*

(Granmo, 2016)

Bayesian Unification of Gradient and Bandit-Based Learning for Accelerated Global Optimisation

> *"However, for continuous optimisation problems or problems with a large number of arms (actions), bandit based approaches are*

hindered by their inability to generalise across arms (typically modelling arms as independent reward sources). This independence assumption leads to slow learning because the expected reward function must be inferred independently for each arm"

". To deal with continuous and large action spaces, several bandit based approaches have recently been proposed that capture interaction among actions. One class of schemes, referred to as global multi-armed bandit schemes, models the expected rewards of the arms as (non-)linear functions of a global parameter γ (Atan et al., 2014). Another family of techniques attacks large action spaces through tree based searching, with X-Armed Bandits finding global maxima when the expected reward (objective) function is "locally Lipschitz" (Bubeck et al., 2010). Finally, Gaussian processes have been applied for smoothing and interpolation, forming the foundation for bandit based exploration and exploitation in continuous action spaces [4]."

(Atan et al., 2014)

Global Multi-armed Bandits with Hölder Continuity

"Standard Multi-Armed Bandit (MAB) problems assume that the arms are independent. However, in many application scenarios, the information obtained by playing an arm provides information about the remainder of the arms. Hence, in such applications, this informativeness can and should be exploited to enable faster convergence to the optimal solution. In this paper, formalize a new class of multi-armed bandit methods, Global Multi-armed Bandit (GMAB), in which arms are globally informative through a global parameter, i.e., choosing an arm reveals information about all the arms. We propose a greedy policy for the GMAB which always selects the arm with the highest estimated expected reward, and prove that it achieves bounded parameter-dependent regret. Hence, this policy selects suboptimal arms only nitely many times, and after a nite number of initial time steps, the optimal arm is selected in all of the remaining time steps with probability one. In addition, we also study how the informativeness of the arms about each other's rewards aects the speed of learning. Specically, we prove that the parameter-free (worst-case) regret is sublinear in time, and decreases with the informativeness of the arms. We also prove a sublinear in time Bayesian risk bound for the GMAB which reduces to the well-known Bayesian risk bound for linearly parameterized bandits when the arms are fully informative. GMABs have applications ranging from drug dosage control to dynamic pricing."

(Bubeck et al., 2010)

X-Armed Bandits

"We consider a generalization of stochastic bandits where the set of arms, X, is allowed to be a generic measurable space and the mean-payoff function is "locally Lipschitz" with respect to a dissimilarity function that is known to the decision maker. Under this condition we construct an arm selection policy, called HOO (hierarchical optimistic optimization), with improved regret bounds compared to previous results for a large class of problems. In particular, our results imply that if X is the unit hypercube in a Euclidean space and the mean-payoff function has a finite number of global maxima around which the behavior of the function is locally continuous with a known smoothness degree, then the expected regret of HOO is bounded up to a logarithmic factor by √n, that is, the rate of growth of the regret is independent of the dimension of the space. We also prove the minimax optimality of our algorithm when the dissimilarity is a metric. Our basic strategy has quadratic computational complexity as a function of the number of time steps and does not rely on the doubling trick. We also introduce a modified strategy, which relies on the doubling trick but runs in linearithmic time. Both results are improvements with respect to previous approaches."

(Saito et al., 2022)

Off-Policy Evaluation for Large Action Spaces via Embeddings

"There are two prominent approaches to deal with large or infinite action spaces in the online bandit literature (Krishnamurthy et al., 2019)Slivkins, 2019). The first one is the parametric approach such as linear or combinatorial bandits, which assumes that the expected reward can be represented as a parametric function of the action such as a linear function (Chu et al., 2011)(Agrawal et al., 2012). There is also a nonparametric approach, which typically makes much weaker assumptions about the rewards, e.g., Lipschitz assumptions. Lipschitz bandits have been studied to address large, structured action spaces such as the [0] 1] interval, where the applications range from dynamic pricing to ad auction. A basic idea in this literature is that similar arms should have similar quality, as per Lipschitz-continuity or some corresponding assumptions on the structure of the action space. The Lipschitz assumption was introduced by (Agrawal, 1995) to the bandit setting. (Kleinberg, 2004) optimally solve this problem in the worst case. (Kleinberg et al., 2013) and Bubeck et al. (2011) rely on the zooming algorithms, which gradually zoom in to the more promising regions of the action space to achieve data-dependent regret bounds. Further works extend this direction by relaxing the assumptions with various local definitions, as well as incorporating contexts into account, as surveyed in Section 4 of Slivkins (2019)."

(Combes et al., 2017)

Minimal Exploration in Structured Stochastic Bandits

"This paper introduces and addresses a wide class of stochastic bandit problems where the function mapping the arm to the corresponding reward exhibits some known structural properties. Most existing structures (e.g. linear, Lipschitz, unimodal, combinatorial, dueling,"

") are covered by our framework. We derive an asymptotic instance-specific regret lower bound for these problems, and develop OSSB, an algorithm whose regret matches this fundamental limit. OSSB is not based on the classical principle of "optimism in the

face of uncertainty" or on Thompson sampling, and rather aims at matching the minimal exploration rates of sub-optimal arms as characterized in the derivation of the regret lower bound. We illustrate the efficiency of OSSB using numerical experiments in the case of the linear bandit problem and show that OSSB outperforms existing algorithms, including Thompson sampling."

Continuum-armed and Lipschitz bandits: definitions and relationship

Continuum-armed bandits are bandits with infinitely many actions, usually in a continuous space like an interval or subset of \(\mathbb{R}^d\). Lipschitz bandits are the structured version where nearby actions must have similar expected rewards, and the modern view treats continuum-armed bandits as a key special case of Lipschitz bandits over metric spaces. (17 sources)

The basic continuum-armed bandit (CAB) setup replaces a finite arm set with a **continuous action domain**. A learner picks an action a_t from a continuum such as an interval or subset of \mathbb{R}^d, and receives a noisy reward whose mean is given by an unknown function f(a_t). This is the setting introduced in early CAB work and it is hard without extra assumptions because there are infinitely many actions to compare. (Gur et al., 2019) (Xu et al., 2022) (Chakraborty et al., 2026)

The standard assumption that makes CAB tractable is **smoothness**, most often a Lipschitz condition: if two actions are close, then their mean rewards must also be close. Several sources make this identification explicit, describing Lipschitz bandits as the model originally introduced as continuum-armed bandits, first on a compact interval and later generalized to broader spaces. (Feng et al., 2021) (Feng et al., 2023) (Liu et al., 2025) (Guo et al., 2025) (Hoi et al., 2018)

A useful way to state the relationship is: **continuum-armed bandits are the geometric domain; Lipschitz bandits add the similarity rule.** In the early form, the arm space was a compact interval, while later work extended the idea to ^d, general Euclidean spaces, and then arbitrary metric or even dissimilarity spaces. Kleinberg et al. formalized this as the **Lipschitz MAB problem**, where strategies form a metric space and the payoff function is Lipschitz with respect to that metric; Bubeck et al. then further generalized to generic measurable arm spaces with local Lipschitzness relative to a known dissimilarity function. (Kleinberg et al., 2008) (Bubeck et al., 2010) (Vakili et al., 2019) (Slivkins et al., 2025)

This shift from “continuous actions” to “metric actions” matters because it shows that the key object is not continuity by itself, but **similarity structure**. Arms can be points in a Euclidean space, a compact interval, or any metric space, as long as the metric tells the learner which actions should have similar rewards. That is why later papers often use “continuum-armed bandit” and “Lipschitz bandit” almost interchangeably in the classical continuous case, while reserving “Lipschitz bandit” for the more general metric-space formulation. (Wang et al., 2019) (Kang et al., 2023) (Slivkins et al., 2025)

The regret picture also helps clarify the relation. For plain Lipschitz or Hölder-continuous reward functions on continuous domains, early results established worst-case rates around T^{(d+1)/(d+2)}, including the notable \Omega(T^{2/3}) lower bound in one-dimensional settings and matching upper bounds. Under stronger conditions beyond bare Lipschitzness, such as favorable local behavior near optima, faster rates like \tilde O(\sqrt{T}) can be achieved. (Feng et al., 2021) (Feng et al., 2023) (Guo et al., 2025) (Potfer et al., 2026) (Vakili et al., 2019) (Bubeck et al., 2010)

Finally, this literature already shows the core contrast with tabular learning. A tabular learner must estimate each action on its own, but in a continuum or metric action space that is impossible or wasteful. The Lipschitz assumption lets the learner use samples from one action to infer bounds for nearby actions, which is why many methods either **discretize the space** or **adaptively refine promising regions** instead of treating every point as unrelated. (Liu et al., 2025) (Magureanu et al., 2014) (Slivkins, 2009)

Evidence

(Gur et al., 2019)

Smoothness-Adaptive Contextual Bandits

"A problem in the literature that shares some similarities with the non-parametric contextual MAB problem is the non-parametric continuum-armed bandit problem (see, e.g., (Agrawal, 1995), where, given a single unknown function f : R d R, at each time t the agent selects an action a t from the continuous action set, and then collects and observes a noisy reward with mean f (a t). While the continuum-armed bandit problem and the contextual bandit problem we consider here are fundamentally different problems, they share the feature that smoothness plays a key role in terms of minimax complexity; see, e.g., Kleinberg (2005), (Auer et al., 2007), and . More recently, Locatelli and Carpentier (2018) studies the problem of adapting to unknown smoothness in a continuum-armed bandit problem in the case where the unknown function f (·) is at most Lipschitz smooth, and shows that under conventional assumptions it is impossible to minimize the cumulative regret rate when the underlying smoothness is unknown."

(Xu et al., 2022)

Towards Agnostic Feature-based Dynamic Pricing: Linear Policies vs Linear Valuation with Unknown Noise

"There exists another family of bandit problem: continuum-armed bandit (CAB) (Agrawal, 1995)(Kleinberg, 2004)(Auer et al., 2007), where the action space is continuum and the reward function is Lipschitz."

(Chakraborty et al., 2026)

Multi-Agent Lipschitz Bandits

"When the set of actions is a continuous domain, such as X = [0, 1] d , the problem becomes intractable without further assumptions, as there are infinitely many arms to explore. The Lipschitz bandit model introduces structural smoothness. The unknown mean-reward function μ : X [0] 1] is assumed to be L-Lipschitz with respect to a norm, typically the Euclidean norm"

"This condition ensures that the mean rewards of nearby points are similar, allowing an algorithm to generalize from a finite number of samples to the entire space. This structure makes the problem tractable, and algorithms for this setting can achieve nearoptimal regret that scales as Õ(T (d+1)/(d+2)), where the exponent depends on the dimension d of the action space."

(Feng et al., 2021)

Lipschitz Bandits With Batched Feedback

"The Lipschitz bandit problem was introduced as "continuum-armed bandits" [3], where the arm space is a compact interval. Along this line, bandits that are Lipschitz (or Hölder) continuous have been studied. For this problem, Kleinberg [27] proves a Ω(T 2/3) lower bound and introduced a matching algorithm"

". For general (doubling) metric spaces, the Zooming bandit algorithm (Kleinberg et al., 2008) and the Hierarchical Optimistic Optimization (HOO) algorithm (Bubeck et al., 2010) were developed. In more recent years, some attention has been focused on Lipschitz bandit problems with certain extra structures"

". Wang et al. (Wang et al., 2019) studied discretization-based Lipschitz bandit algorithms from a Gaussian process perspective"

". Contextual Lipschitz bandits have been studied by Slivkins (Slivkins, 2009). Contextual bandits with continuous actions have also been studied by Krishnamurthy et al. (Krishnamurthy et al., 2019) and Majzoubi et al. (Majzoubi et al., 2020) through a smoothness approach"

". Esfandiari et al. (Esfandiari et al., 2019) studies batched linear bandits and batched adversarial bandits"

". Li and Scarlett (Li et al., 2021) studies batched Gaussian process bandits."

(Feng et al., 2023)

A Lipschitz Bandits Approach for Continuous Hyperparameter Optimization

"The Lipschitz bandit problem was introduced as "continuum-armed bandits" (Agrawal, 1995), where the arm space is a compact interval. For this problem, (Kleinberg, 2004) proved an Ω(T 2/3) lower bound and introduced an algorithm that matches this lower bound. Under extra conditions on top of Lipschitzness, regret rate of O(T 1/2) was achieved (Auer et al., 2007)(Cope, 2009). For compact doubling metric spaces, the Zooming bandit algorithm (Kleinberg et al., 2008), the Hierarchical Optimistic Optimization (HOO) algorithm (Bubeck et al., 2010), and the BLiN algorithm (Feng et al., 2021) were developed. Additionally, some attention has been focused on Lipschitz bandit problems where extra conditions are imposed. To name a few, (Bubeck et al., 2011) study Lipschitz bandits for differentiable rewards, which enables algorithms to run without explicitly knowing the Lipschitz constants. Wang et al. (2020) studied discretization-based Lipschitz bandit algorithms from a Gaussian process perspective. (Magureanu et al., 2014) derive a new concentration inequality and study discrete Lipschitz bandits. The idea of robust mean estimators (Bickel, 1965) (Alon et al., 1996)(Bubeck et al., 2012) was applied to the Lipschitz bandit problem to cope with heavy-tail rewards, leading to the development of a near-optimal algorithm for Lipschitz bandit with heavy-tailed rewards (Lu et al., 2019). Lipschitz bandits where a clustering is used to infer the underlying metric, has been studied by (Wanigasekara et al., 2019). Contextual Lipschitz bandits have also been studied (Slivkins, 2009)(Krishnamurthy et al., 2019)."

(Liu et al., 2025)

Lipschitz Bandits with Stochastic Delayed Feedback

"The Lipschitz bandit problem, also known as continuum-armed bandits, was first introduced in (Agrawal, 1995). The problem was then extended to the general metric space in (Kleinberg et al., 2008). A common approach to solving Lipschitz bandit problems is to discretize the continuous arm spac, either uniformly or adaptively, and then apply standard finite-armed bandit algorithms to deal with the exploration-exploitation tradeoff. This reduction enables the use of strategies such as the Upper Confidence Bound (UCB) method (Kleinberg et al., 2013), Thompson Sampling (TS) (Vernade et al., 2018) and elimination-based techniques (Feng et al., 2021)."

(Guo et al., 2025)

Batched Bandits with Heavy-Tailed Rewards

"The Lipschitz bandit problem was originally introduced as "continuum-armed bandits" [3], where the arm set is a compact interval. Subsequent work extended this setting to bandits with Lipschitz (or Hölder) continuous reward functions. (Kleinberg, 2004) established a Ω(T 2/3) lower bound for this problem and proposed an algorithm that matches it. When additional conditions beyond Lipschitzness are imposed, a regret rate of O(T 1/2) becomes achievable [10]20]."

(Hoi et al., 2018)

Online Learning: A Comprehensive Survey

"Among these studies, one niche sub-topic is continuum-armed bandits (Kleinberg, 2004)Chowdhury and Gopalan, 2017), where the arms lie in some Euclidean (or metric) space and their mean-reward is a deterministic and smooth (e.g., Lipschitz) function of the arms, a.k.a. Lipschitz Bandit (Magureanu et al., 2014)."

(Kleinberg et al., 2008)

Multi-armed bandits in metric spaces

"In a multi-armed bandit problem, an online algorithm chooses from a set of strategies in a sequence of n trials so as to maximize the total payoff of the chosen strategies. While the performance of bandit algorithms with a small finite strategy set is quite well understood, bandit problems with large strategy sets are still a topic of very active investigation, motivated by practical applications such as online auctions and web advertisement. The goal of such research is to identify broad and natural classes of strategy sets and payoff functions which enable the design of efficient solutions. In this work we study a very general setting for the multi-armed bandit problem in which the strategies form a metric space, and the payoff function satisfies a Lipschitz condition with respect to the metric. We refer to this problem as the "Lipschitz MAB problem". We present a complete solution for the multi-armed problem in this setting. That is, for every metric space (L,X) we define an isometry invariant Max Min COV(X) which bounds from below the performance of Lipschitz MAB algorithms for X, and we present an algorithm which comes arbitrarily close to meeting this bound. Furthermore, our technique gives even better results for benign payoff functions."

(Bubeck et al., 2010)

X-Armed Bandits

"We consider a generalization of stochastic bandits where the set of arms, X, is allowed to be a generic measurable space and the mean-payoff function is "locally Lipschitz" with respect to a dissimilarity function that is known to the decision maker. Under this condition we construct an arm selection policy, called HOO (hierarchical optimistic optimization), with improved regret bounds compared to previous results for a large class of problems. In particular, our results imply that if X is the unit hypercube in a Euclidean space and the mean-payoff function has a finite number of global maxima around which the behavior of the function is locally continuous with a known smoothness degree, then the expected regret of HOO is bounded up to a logarithmic factor by √n, that is, the rate of growth of the regret is independent of the dimension of the space. We also prove the minimax optimality of our algorithm when the dissimilarity is a metric. Our basic strategy has quadratic computational complexity as a function of the number of time steps and does not rely on the doubling trick. We also introduce a modified strategy, which relies on the doubling trick but runs in linearithmic time. Both results are improvements with respect to previous approaches."

(Vakili et al., 2019)

Stochastic Gradient Descent on a Tree: an Adaptive and Robust Approach to Stochastic Convex Optimization

"The continuum armed bandit under Lipschitz assumption (not necessarily convex) has been studied in (Agrawal, 1995)(Kleinberg, 2004)(Kleinberg et al., 2008) where higher orders of regret were shown. The X -armed bandit introduced in (Bubeck et al., 2010) considered a Lipschitz function with respect to a dissimilarity function known to the learner. Under the assumption of a finite number of global optima and a particular smoothness property, an Õ(✓ T) regret was shown. While the proposed policy in (Bubeck et al., 2010) uses a tree structure for updating the indexes in a bandit algorithm, it is fundamentally different from RWT in that the policy does not induce a random walk on the tree."

(Slivkins et al., 2025)

Greedy Algorithm for Structured Bandits: A Sharp Characterization of Asymptotic Success / Failure

"Lipschitz bandits is a special case of StructuredMAB in which all reward functions f F satisfy Lipschitz condition, |f (a) − f (a ')| ≤

D(a, a '), for any two arms a, a ' ∈ A and some known metric D on A. Introduced in (Kleinberg et al., 2008); (Bubeck et al., 2008), Lipschitz bandits have been studied extensively since then, see Slivkins (2019, Ch. 4.4) for a survey. The paradigmatic case is continuum-armed bandits (Agrawal, 1995)(Kleinberg, 2004)(Auer et al., 2007), where one has action"

(Wang et al., 2019)

A Practical Bandit Method with Advantages in Neural Network Tuning

"One recent line of work on stochastic bandit problems considers the case where the arm space is infinite. In this setting, the arms are usually assumed to be in a subset of the Euclidean space (or a more general metric space), and the expected payoff function is assumed to be a function of the arms. Some works along this line model the expected payoff as a linear function of the arms (Auer, 2003)(Dani et al., 2008)(Li et al., 2010)(Abbasi-Yadkori et al., 2011)(Agrawal et al., 2012)(Abeille et al., 2016); some algorithms model the expected payoff as Gaussian processes over the arms (Srinivas et al., 2009;(Contal et al., 2013)(Freitas et al., 2012)Vazquez and Bect, 2007); some algorithms assume that the expected payoff is a Lipschitz function of the arms (Slivkins, 2009)(Kleinberg et al., 2008)Bubeck et al., 2011); and some assume locally Hölder payoffs on the real line (Auer et al., 2007). When the arms are continuous and equipped with a metric, and the expected payoff is Lipschitz continuous in the arm space, we refer to the problem as a stochastic Lipschitz bandit problem. In addition, when the agent's decisions are made with the aid of contextual information, we refer to the problem as a contextual stochastic Lipschitz bandit problem."

(Kang et al., 2023)

Robust Lipschitz Bandits to Adversarial Corruptions

"Lipschitz bandit is a variant of stochastic bandits that deals with a continuous arm set defined on a metric space, where the reward function is subject to a Lipschitz constraint"

". Extensive studies have been conducted on MAB and its variations, including linear bandit [1], matrix bandit [20], etc."

(Potfer et al., 2026)

Instance-dependent Stochastic Lipschitz bandit

"Lipschitz bandit algorithms are a natural tool for sequential optimization over continuous action spaces"

". In the worst case, R T scales as Θ T d+1 d+2

for Lipschitz/continuum-armed bandits (Auer et al., 2007;Kleinberg, 2004)."

(Magureanu et al., 2014)

Lipschitz Bandits: Regret Lower Bound and Optimal Algorithms

"We consider stochastic multi-armed bandit problems where the expected reward is a Lipschitzfunction of the arm, and where the set of arms is either discrete or continuous. For discrete Lipschitzbandits, we derive asymptotic problem specific lower bounds for the regret satisfied by anyalgorithm, and propose OSLB and CKL-UCB, two algorithms that efficiently exploit the Lipschitzstructure of the problem. In fact, we prove that OSLB is asymptotically optimal, as its asymptoticregret matches the lower bound"

".For continuous Lipschitz bandits, we propose to first discretize the actionspace, and then apply OSLB or CKL-UCB, algorithms that provably exploit the structure efficiently"

".Finally the results and algorithms areextended to contextual bandits with similarities"

".(e) We extend our results and algorithms to the case of contextual bandits with similarities as investigated in Slivkins (2011)."

(Slivkins, 2009)

Contextual Bandits with Similarity Information

"In a multi-armed bandit (MAB) problem, an online algorithm makes a sequence of choices. In each round it chooses from a time-invariant set of alternatives and receives the payoff associated with this alternative. While the case of small strategy sets is by now well-understood, a lot of recent work has focused on MAB problems with exponentially or infinitely large strategy sets, where one needs to assume extra structure in order to make the problem tractable. In particular, recent literature considered information on similarity between arms.

We consider similarity information in the setting of contextual bandits, a natural extension of the basic MAB problem where before each round an algorithm is given the context--a hint about the payoffs in this round. Contextual bandits are directly motivated by placing advertisements on web pages, one of the crucial problems in sponsored search. A particularly simple way to represent similarity information in the contextual bandit setting is via a similarity distance between the context-arm pairs which bounds from above the difference between the respective expected payoffs.

Prior work on contextual bandits with similarity uses "uniform" partitions of the similarity space, so that each context-arm pair is approximated by the closest pair in the partition. Algorithms based on "uniform" partitions disregard the structure of the payoffs and the context arrivals, which is potentially wasteful. We present algorithms that are based on adaptive partitions, and take advantage of "benign" payoffs and context arrivals without sacrificing the worst-case performance. The central idea is to maintain a finer partition in high-payoff regions of the similarity space and in popular regions of the context space. Our results apply to several other settings, e.g., MAB with constrained temporal change (Slivkins and Upfal, 2008) and sleeping bandits (Kleinberg et al., 2008a)."

Lipschitz/metric-space algorithms and regret guarantees

The main algorithmic idea in Lipschitz bandits is to avoid treating the action space as a flat table. Instead, algorithms either discretize the space or adaptively refine promising regions, and the best regret guarantees depend on the metric geometry of near-optimal actions rather than just on how many nominal actions there are. (16 sources)

Algorithmically, the literature splits into two broad approaches. The simplest is **fixed discretization**: replace the continuous action set by a grid of K points, run a standard finite-arm method on that grid, and balance learning error against discretization error. In one dimension, this gives the familiar tradeoff \sqrt{TK} from bandit learning plus about T/K from approximation, which is optimized at K \asymp T^{1/3} and yields regret of order T^{2/3}; this rate is known to be optimal in the basic Lipschitz setting. This discretize-then-banditize view appears both in classical CAB work and in later structured methods such as OSLB and CKL-UCB for discrete/continuous Lipschitz problems. (Jalota et al., 2023) (Slivkins, 2009) (Magureanu et al., 2014)

A more powerful line of work uses **adaptive discretization**. Here the learner does not commit to a uniform grid in advance; it starts coarse and refines only regions that look promising. The notable examples are the **zooming algorithm**, which allocates more resolution near high-reward areas, and **HOO** (Hierarchical Optimistic Optimization), which explores a tree of nested regions and expands nodes optimistically near candidate maximizers. These methods were among the first to get near-optimal worst-case guarantees up to log factors while also improving on benign instances. (Liu et al., 2024) (Kleinberg et al., 2008) (Kleinberg et al., 2013) (Bubeck et al., 2011) (Zhou et al., 2025)

The key regret message is that performance is governed by **intrinsic metric complexity**, not by raw cardinality. In the worst case, classical Lipschitz rates scale like T^{(d+1)/(d+2)} on d-dimensional spaces, but zooming-style analyses sharpen this by replacing ambient dimension with a finer notion such as the **zooming dimension** of the near-optimal region. For zooming dimension d_z, the optimal rate becomes \widetilde O(T^{(d_z+1)/(d_z+2)}), with matching lower-bound intuition and later variants preserving the same scaling in batched settings. (Feng et al., 2021) (Sharma et al., 2026) (Kleinberg et al., 2008) (Kleinberg et al., 2013)

This is also where metric-space bandits differ most clearly from tabular learners. A tabular method must sample each arm as if it were unrelated to the others, while zooming and tree-based methods can rule out or prioritize **whole regions** using the metric and confidence bounds. Slivkins emphasizes that adaptive partitions can focus on high-payoff and frequently visited parts of the similarity space, rather than wasting effort on uniform partitions; Krishnamurthy et al. describe this as “zooming” on promising action regions under weak continuity assumptions. (Slivkins, 2009) (Krishnamurthy et al., 2019)

Another theme is **adaptivity**: can the learner get the right rate without knowing smoothness constants or time horizon in advance? For globally Lipschitz rewards on ^d, Bubeck et al. showed minimax-optimal rates can be achieved without prior knowledge of the Lipschitz constant L or T. But adaptivity is not always free: later work notes lower bounds showing that certain adaptive guarantees in Lipschitz bandits require side information such as knowledge of a smoothness scale L_\star, and there is a real price of adaptivity when smoothness is unknown. Related methods such as StoSOO and later optimistic optimization variants handle weaker **local smoothness** assumptions and try to adapt to unknown regularity. (Bubeck et al., 2011) (Foster et al., 2020) (Krishnamurthy et al., 2019) (Valko et al., 2013)

The framework has also broadened beyond the original stochastic, stationary case. There are now versions for **contextual Lipschitz bandits**, **adversarial Lipschitz bandits**, **batched feedback**, and **non-stationary**

Lipschitz bandits, often reusing the same core idea of hierarchical or adaptive discretization. This makes Lipschitz/metric-space learning a general template for structured exploration: use geometry to share evidence across nearby actions, then spend samples where the reward landscape appears most competitive. (Podimata et al., 2020) (Feng et al., 2021) (Nguyen et al., 2025) (Yi et al., 2025) (Liu et al., 2024)

Finally, there are also **discrete Lipschitz bandits**, where the arm set is finite but rewards obey a known graph or metric structure. In that setting, one can derive **problem-dependent lower bounds** and design asymptotically optimal algorithms such as OSLB that exploit the Lipschitz coupling among arms instead of ignoring it. This is a useful bridge between pure tabular learning and continuous-action models: even with finitely many arms, structure can still let one arm teach the learner about several others. (Magureanu et al., 2014)

Evidence

(Jalota et al., 2023)

Online Learning for Equilibrium Pricing in Markets under Incomplete Information

> *"For instance, in Lipschitz bandits (without contexts) where the action space is continuous and the reward function is Lipschitz in the actions, after restricting to a set of K uniformly spaced actions, the regret incurred by a bandit algorithm, e.g., EXP3, is on the order of ✓ T K and the discretization error is on the order of T K . After picking the optimal choice of K = T 1/3 , the overall regret for a one-dimensional Lipschitz bandit is on the order of T 2/3 , which is known to be optimal for the Lipschitz bandit setting [31](Slivkins, 2009)"*
>
> *".Recall that, as previously discussed, the optimal regret for the Lipschitz bandit problem is on the order of T 2/3 . Therefore, the O(T 2/3) regret established in Theorem 3 for finite function classes (see Remark 3) is consistent with known lower bounds in bandit settings with infinite action spaces, indicating that our result matches the expected regret scaling in such settings."*

(Slivkins, 2009)

Contextual Bandits with Similarity Information

> *"In a multi-armed bandit (MAB) problem, an online algorithm makes a sequence of choices. In each round it chooses from a time-invariant set of alternatives and receives the payoff associated with this alternative. While the case of small strategy sets is by now well-understood, a lot of recent work has focused on MAB problems with exponentially or infinitely large strategy sets, where one needs to assume extra structure in order to make the problem tractable. In particular, recent literature considered information on similarity between arms.*
>
> *We consider similarity information in the setting of contextual bandits, a natural extension of the basic MAB problem where before each round an algorithm is given the context--a hint about the payoffs in this round. Contextual bandits are directly motivated by placing advertisements on web pages, one of the crucial problems in sponsored search. A particularly simple way to represent similarity information in the contextual bandit setting is via a similarity distance between the context-arm pairs which bounds from above the difference between the respective expected payoffs.*
>
> *Prior work on contextual bandits with similarity uses "uniform" partitions of the similarity space, so that each context-arm pair is approximated by the closest pair in the partition. Algorithms based on "uniform" partitions disregard the structure of the payoffs and the context arrivals, which is potentially wasteful. We present algorithms that are based on adaptive partitions, and take advantage of "benign" payoffs and context arrivals without sacrificing the worst-case performance. The central idea is to maintain a finer partition in high-payoff regions of the similarity space and in popular regions of the context space. Our results apply to several other settings, e.g., MAB with constrained temporal change (Slivkins and Upfal, 2008) and sleeping bandits (Kleinberg et al., 2008a)."*

(Magureanu et al., 2014)

Lipschitz Bandits: Regret Lower Bound and Optimal Algorithms

> *"We consider stochastic multi-armed bandit problems where the expected reward is a Lipschitzfunction of the arm, and where the set of arms is either discrete or continuous. For discrete Lipschitzbandits, we derive asymptotic problem specific lower bounds for the regret satisfied by anyalgorithm, and propose OSLB and CKL-UCB, two algorithms that efficiently exploit the Lipschitzstructure of the problem. In fact, we prove that OSLB is asymptotically optimal, as its asymptoticregret matches the lower bound"*
>
> *".For continuous Lipschitz bandits, we propose to first discretize the actionspace, and then apply OSLB or CKL-UCB, algorithms that provably exploit the structure efficiently"*
>
> *".Finally the results and algorithms areextended to contextual bandits with similarities"*

".(e) We extend our results and algorithms to the case of contextual bandits with similarities as investigated in Slivkins (2011)."

(Liu et al., 2024)

Batched Stochastic Bandit for Nondegenerate Functions

"Throughout the years, multiple variations of the stochastic MAB problems have been intensively investigated, including linear bandits (Auer, 2003)(Dani et al., 2007)(Chu et al., 2011)(Abbasi-Yadkori et al., 2011), Gaussian process bandits (Srinivas et al., 2009) (Contal et al., 2013), bandits in metric spaces (Kleinberg, 2004)(Kleinberg et al., 2008)(Bubeck et al., 2008)23,(Podimata et al., 2020), just to name a few"

".In its early stage, bandits in metric spaces primarily focus on bandit learning over [0, 1] [(Agrawal, 1995)(Kleinberg, 2004)(Auer et al., 2007)(Cope, 2009). Afterwards, algorithms for bandits over more general metric spaces were developed (Kleinberg et al., 2008) (Bubeck et al., 2008)23,(Bubeck et al., 2011)(Magureanu et al., 2014)(Lu et al., 2019)(Krishnamurthy et al., 2019)(Majzoubi et al., 2020)(Feng et al., 2021). In particular, the Zooming bandit algorithm (Kleinberg et al., 2008)(Slivkins, 2009) and the Hierarchical Optimistic Optimization (HOO) algorithm (Bubeck et al., 2008)[23] were the first algorithms that optimally solve the Lipschitz bandit problem (up to logarithmic factors). Subsequently, (Valko et al., 2013) considered an early version of nondegenerate functions, and built its connection to Lipschitz bandits (Kleinberg et al., 2008)(Bubeck et al., 2008)[23]. (Valko et al., 2013) proposed StoSOO algorithm for pure exploration of function that is locally smooth with respect to some semimetric. But, to our knowledge, bandit problems with such functions have not been explored."

(Kleinberg et al., 2008)

Multi-armed bandits in metric spaces

"In a multi-armed bandit problem, an online algorithm chooses from a set of strategies in a sequence of n trials so as to maximize the total payoff of the chosen strategies. While the performance of bandit algorithms with a small finite strategy set is quite well understood, bandit problems with large strategy sets are still a topic of very active investigation, motivated by practical applications such as online auctions and web advertisement. The goal of such research is to identify broad and natural classes of strategy sets and payoff functions which enable the design of efficient solutions. In this work we study a very general setting for the multi-armed bandit problem in which the strategies form a metric space, and the payoff function satisfies a Lipschitz condition with respect to the metric. We refer to this problem as the "Lipschitz MAB problem". We present a complete solution for the multi-armed problem in this setting. That is, for every metric space (L,X) we define an isometry invariant Max Min COV(X) which bounds from below the performance of Lipschitz MAB algorithms for X, and we present an algorithm which comes arbitrarily close to meeting this bound. Furthermore, our technique gives even better results for benign payoff functions."

(Kleinberg et al., 2013)

Bandits and Experts in Metric Spaces

"In a multi-armed bandit problem, an online algorithm chooses from a set of strategies in a sequence of trials to maximize the total payoff of the chosen strategies. While the performance of bandit algorithms with a small finite strategy set is well understood, bandit problems with large strategy sets are still a topic of active investigation, motivated by practical applications, such as online auctions and web advertisement. The goal of such research is to identify broad and natural classes of strategy sets and payoff functions that enable the design of efficient solutions. In this work, we study a general setting for the multi-armed bandit problem, in which the strategies form a metric space, and the payoff function satisfies a Lipschitz condition with respect to the metric. We refer to this problem as the Lipschitz MAB problem. We present a solution for the multi-armed bandit problem in this setting. That is, for every metric space, we define an isometry invariant that bounds from below the performance of Lipschitz MAB algorithms for this metric space, and we present an algorithm that comes arbitrarily close to meeting this bound. Furthermore, our technique gives even better results for benign payoff functions. We also address the full-feedback ("best expert") version of the problem, where after every round the payoffs from all arms are revealed."

(Bubeck et al., 2011)

Lipschitz Bandits without the Lipschitz Constant

"We consider the setting of stochastic bandit problems with a continuum of arms indexed by [0, 1]d. We first point out that the strategies considered so far in the literature only provided theoretical guarantees of the form: given some tuning parameters, the regret is small with respect to a class of environments that depends on these parameters. This is however not the right perspective, as it is the strategy that should adapt to the specific bandit environment at hand, and not the other way round. Put differently, an adaptation issue is raised. We solve it for the special case of environments whose mean-payoff functions are globally Lipschitz. More precisely, we show that the minimax optimal orders of magnitude Ld/(d+2) T(d+1)/(d+2) of the regret bound over T time instances against an environment whose mean-payoff function f is Lipschitz with constant L can be achieved without knowing L or T in advance. This is in contrast to all previously known strategies, which require to some extent the knowledge of L to achieve this performance guarantee."

(Zhou et al., 2025)

Representative Action Selection for Large Action Space Bandit Families

"Lipschitz Bandits study bandit problems over continuous or very large action spaces, under the assumption that the reward function varies smoothly with respect to a metric on the action space (Kleinberg et al., 2008)(Slivkins, 2009). Common approaches include the Zooming algorithm (Kleinberg et al., 2008), which refines the exploration process according to the local geometry of nearoptimal regions rather than relying on uniform discretization, achieving regret tied to the "zooming dimension"; and Hierarchical Optimistic Optimization [Bubeck et al., 2011], which maintains a binary tree whose nodes correspond to measurable regions of the action space, explores by traversing the tree according to optimistic values, and progressively expands nodes to refine the partition in near-optimal regions. This line of work is closely related to ours, as both deal with large action spaces endowed with a similarity structure, and the Zooming algorithm likewise exploits the notion of □ nets"

". Srinivas et al. [2009] model the reward function as a sample from a Gaussian process prior and propose the GP-UCB algorithm, which achieves cumulative regret O T γ T log |A full | for finite action spaces, where γ T denotes the maximum information gain after T rounds. The widespread adoption of this approach-both for finite action sets [Valko et al., 2013, Li andScarlett, 2022] and continuous domains [Chowdhury and Gopalan, 2017] highlights the practicality of assuming that action outcomes are correlated"

". In many natural situations, while the nominal set of actions is large, actions are highly correlated: many yield similar rewards across environments, making it wasteful to maintain the full set."

(Feng et al., 2021)

Lipschitz Bandits With Batched Feedback

"The Lipschitz bandit problem was introduced as "continuum-armed bandits" [3], where the arm space is a compact interval. Along this line, bandits that are Lipschitz (or Hölder) continuous have been studied. For this problem, Kleinberg [27] proves a Ω(T 2/3) lower bound and introduced a matching algorithm"

". For general (doubling) metric spaces, the Zooming bandit algorithm (Kleinberg et al., 2008) and the Hierarchical Optimistic Optimization (HOO) algorithm (Bubeck et al., 2010) were developed. In more recent years, some attention has been focused on Lipschitz bandit problems with certain extra structures"

". Wang et al. (Wang et al., 2019) studied discretization-based Lipschitz bandit algorithms from a Gaussian process perspective"

". Contextual Lipschitz bandits have been studied by Slivkins (Slivkins, 2009). Contextual bandits with continuous actions have also been studied by Krishnamurthy et al. (Krishnamurthy et al., 2019) and Majzoubi et al. (Majzoubi et al., 2020) through a smoothness approach"

". Esfandiari et al. (Esfandiari et al., 2019) studies batched linear bandits and batched adversarial bandits"

". Li and Scarlett (Li et al., 2021) studies batched Gaussian process bandits."

(Sharma et al., 2026)

Lipschitz Dueling Bandits over Continuous Action Spaces

"A separate line of work studies bandits over continuous action spaces under Lipschitz assumptions, where the learner observes noisy scalar rewards (Kleinberg et al., 2013)(Zhu et al., 2025)(Kleinberg, 2004). The problem was first introduced by (Kleinberg, 2004), which proposed a discretization-based algorithm for 1-dimensional functions. Later, (Kleinberg et al., 2008) presented the zooming algorithm for general d-dimensional Lipschitz functions and established a lower bound of Ω(T dz +1 dz +2), where d z is the zooming dimension of the near-optimal region. The zooming algorithm adaptively refines the most promising regions of the action space. Subsequent work has proposed alternative approaches to exploit the structure of the reward function. In particular, (Bubeck et al., 2011) introduced the Hierarchical Optimistic Optimization (HOO) algorithm, which leverages a tree-based partitioning of the space and focuses exploration near the maxima. More recently, there has been increasing interest in improving the space efficiency of such methods. In this direction, (Zhu et al., 2025) proposed a recursive variant of the zooming algorithm with O(log T) space complexity"

". there is a rich line of work that has investigated bandit problems with structured action spaces. These either include convex (Lattimore & Gyorgy, 2021) or a more general Lipschitz action space (Zhu & Huang, 2025;Kleinberg, 2004). In particular, stochastic convex bandits assume that rewards are generated by a convex function over a convex set, while the Lipschitz bandits assume that rewards are generated by an unknown Lipschitz function over a continuous domain, allowing algorithms to exploit geometric structure and achieve regret guarantees that depend on an intrinsic notion of dimension rather than the cardinality of the action space. The best achievable regret in such spaces are O(d 4.5 ✓ T polylog(T)) in stochastic convex bandits with d being the dimension of the space (Lattimore & Gyorgy, 2021) and O(T dz +1 dz +2) in Lipschitz bandits, with d z being the zooming dimension of the near optimal region (Zhu & Huang, 2025)."

(Krishnamurthy et al., 2019)

Contextual Bandits with Continuous Actions: Smoothing, Zooming, and Adapting

"The parametric approach, including work on linear or combinatorial bandits, posits that the loss is a parametric function of the action, e.g., a linear function (c.f., Lattimore and Szepesvári (2018); Bubeck et al. (2012) for surveys). The nonparametric approach,

which is closer to our results,7 typically makes much weaker continuity assumptions"

".Bandits with Lipschitz assumptions were introduced in Agrawal (1995), and optimally solved in the worst case by (Kleinberg, 2004). (Kleinberg et al., 2008)(Kleinberg et al., 2013); Bubeck et al. (2011a) achieve data-dependent regret bounds via algorithms that "zoom in" on the more promising regions of the action space. Several papers relax global smoothness assumptions with various local definitions (Auer et al., 2007)(Kleinberg et al., 2008)(Kleinberg et al., 2013)Bubeck et al., 2011a;(Slivkins, 2011)(Minsker, 2013) (Grill et al., 2015). While the assumptions vary, our smoothing-based approach can be used in many of these settings"

".(Slivkins, 2009) posits that the mean loss function is Lipschitz in both context x and action a, and the learner must compete with the best mapping from X to A. Both algorithm and guarantees exhibit "zooming" behavior in the action space, which is qualitatively similar to ours."

(Foster et al., 2020)

Open Problem: Model Selection for Contextual Bandits

"Locatelli and Carpentier (2018) also observe that knowledge of L can enable adaptive guarantees in Lipschitz bandits, where adaptivity is not possible in the absence of such information"

"Related lower bounds are also known for Lipschitz bandits (Locatelli and Carpentier, 2018;Krishnamurthy et al., 2019). On the positive side, (Chatterji et al., 2019) show that, with distributional assumptions, it is possible to adapt between multi-armed bandits and linear contextual bandits."

(Valko et al., 2013)

Stochastic Simultaneous Optimistic Optimization

"We study the problem of global maximization of a function f given a finite number of evaluations perturbed by noise. We consider a very weak assumption on the function, namely that it is locally smooth (in some precise sense) with respect to some semi-metric, around one of its global maxima. Compared to previous works on bandits in general spaces (Kleinberg et al., 2008; Bubeck et al., 2011a) our algorithm does not require the knowledge of this semi-metric. Our algorithm, StoSOO, follows an optimistic strategy to iteratively construct upper confidence bounds over the hierarchical partitions of the function domain to decide which point to sample next. A finite-time analysis of StoSOO shows that it performs almost as well as the best specifically-tuned algorithms even though the local smoothness of the function is not known."

(Podimata et al., 2020)

Adaptive Discretization against an Adversary: Lipschitz bandits, Dynamic Pricing, and Auction Tuning

"Lipschitz bandits is a prominent version of multi-armed bandits that studies large, structured action spaces such as the $[0,1]$ interval, where similar actions are guaranteed to have similar rewards. A central theme here is the adaptive discretization of the action space, which gradually ``zooms in" on the more promising regions thereof. The goal is to take advantage of ``nicer" problem instances, while retaining near-optimal worst-case performance. While the stochastic version of the problem is well-understood, the general version with adversarial rewards is not. We provide the first algorithm (\emph{Adversarial Zooming}) for adaptive discretization in the adversarial version, and derive instance-dependent regret bounds. In particular, we recover the worst-case optimal regret bound for the adversarial version, and the instance-dependent regret bound for the stochastic version. We apply our algorithm to several fundamental applications -- including dynamic pricing and auction reserve tuning -- all under adversarial reward models. While these domains often violate Lipschitzness, our analysis only requires a weaker version thereof, allowing for meaningful regret bounds without additional smoothness assumptions. Notably, we extend our results to multi-product dynamic pricing with non-smooth reward structures, a setting which does not even satisfy one-sided Lipschitzness."

(Nguyen et al., 2025)

Non-Stationary Lipschitz Bandits

"We study the problem of non-stationary Lipschitz bandits, where the number of actions is infinite and the reward function, satisfying a Lipschitz assumption, can change arbitrarily over time"

". We introduced MDBE, a novel algorithm that leverages a hierarchical discretization of the action space to exploit the Lipschitz structure of the evolving reward function"

". Additionally, extending the framework of significant shift detection to other structured bandit settings, such as linear or convex bandits, is an exciting avenue for future research."

(Yi et al., 2025)

Quantum Lipschitz Bandits

"Lipschitz bandits, along with other notable extensions of the multi-armed bandit framework such as dueling bandits (Yue et al., 2012)Yi, Kang, and Li 2024;Verma et al. 2024), kernelized bandits (Ding et al., 2021)(Dai et al., 2023)(Chowdhury et al., 2017), and

parametric bandits (Ganguly et al., 2023)(Abbasi-Yadkori et al., 2011)(Kang et al., 2024)Kang et al. 2025), have been extensively explored to tackle increasingly complex problem settings. Most prior research on the stochastic Lipschitz bandits builds on two primary approaches. The first involves uniformly discretizing the action space into a fixed grid, enabling the application of any standard multi-armed bandit algorithms (Kleinberg, 2004)(Lu et al., 2019). The second approach emphasizes dynamic discretization, where more arms are sampled in promising regions of the action space. In this approach, classical methods such as Upper Confidence Bound (UCB) (Bubeck et al., 2008)(Kleinberg et al., 2013)(Lu et al., 2019), Thompson Sampling (TS) (Kang, Hsieh, and Lee 2024), and elimination method (Feng et al., 2021)) can be utilized to balance the exploration-exploitation tradeoff"

". Beyond these works, various extensions of Lipschitz bandits have been thoroughly explored, including contextual Lipschitz bandits (Slivkins 2011a), non-stationary Lipschitz bandits (Kang, Hsieh, and Lee 2024), and taxonomy bandits (Slivkins 2011b)."

Structured bandits as a unifying framework

Structured bandits give a common language for bandit problems where arms are related rather than independent. The usual formal idea is that all arm means are tied together by known reward maps and one unknown parameter, so pulling one arm can teach the learner about many others. (12 sources)

A useful unifying view is that **structured bandits are bandits with side information or constraints on how arm means fit together**. Jun et al. describe them broadly as problems where there are "pieces of side information or constraints in (and among) the mean rewards," with the ordinary K-armed bandit as a degenerate special case but the term usually reserved for nontrivial structure (Jun et al., 2020). In one common formalization, there is a fixed but unknown parameter \theta, and each arm k has a known mean-reward map \mu_k(\theta); then learning from one arm updates beliefs about the rewards of others through the shared \theta (Gupta et al., 2019) (Gupta et al., 2021). This is the cleanest statement of how structured learners differ from tabular ones: **tabular learners estimate each arm separately, while structured learners infer a common reward mechanism** (Gupta et al., 2019) (Zhang, 2024).

Seen this way, many familiar models are just special cases inside one larger family. Xu et al. explicitly note that a general structured-bandit formulation can include **Lipschitz bandits, unimodal bandits, linear bandits**, and related models as special cases (Xu et al., 2018). Gupta et al. make the same point for **linear, generalized linear, Lipschitz, global, and regional bandits** (Gupta et al., 2019), while Gupta et al. (2021) also list **global bandits, regional bandits, and generalized linear bandits** as special cases of the hidden-parameter view (Gupta et al., 2021). This perspective puts earlier strands in one map: metric smoothness gives local sharing in Lipschitz bandits (Xu et al., 2018) (Kleinberg et al., 2008), graph or order structure gives sharing in unimodal bandits (Xu et al., 2018) (Combes et al., 2014), and low-dimensional parameterizations give global sharing in linear and related models (Jun et al., 2020) (Rusmevichientong et al., 2008).

The framework also helps compare **how much information a pull reveals**. In **global bandits**, one arm can be informative about all arms through a single global parameter, and more informativeness leads to faster learning (Gupta et al., 2019) (Atan et al., 2014). In **regional bandits**, this sharing is partial: arms are split into groups, and pulling one arm reveals information mainly about other arms in the same group, which bridges the gap between non-informative tabular bandits and fully global models (Gupta et al., 2019) (Wang et al., 2018). In **unimodal** or graph-structured settings, the learner can use the assumed shape of rewards over a graph or partial order to avoid exploring every arm uniformly, and regret can even avoid direct dependence on the total number of arms (Xu et al., 2018) (Combes et al., 2014).

Another benefit of the unifying view is that it connects **regret minimization** and **best-arm identification**. Structured best-arm work in linear bandits shows that one should sample arms that are most informative about the common structure, not only arms that currently look best (Gupta et al., 2021) (Soare et al., 2014). More general structured best-arm formulations extend this logic beyond linearity: the sample complexity depends on how the shared structure constrains plausible reward vectors, not just on estimating each mean one by one (Gupta et al., 2021) (Huang et al., 2017). That is the main conceptual gain of structured bandits as a framework: it lets us ask **what observations are most informative about the whole reward system**, rather than only which arm to try next.

So, as a synthesis, **structured bandits unify the main non-tabular ideas by replacing "one unknown mean per**

arm" with "one structured model over all arms." The exact structure may come from a metric, a graph, a shape constraint, a linear feature map, or grouped/global parameters, but the goal is the same: let evidence from one action generalize to similar or linked actions, and thereby learn much faster than a tabular method that treats every arm as unrelated (Jun et al., 2020) (Gupta et al., 2019) (Gupta et al., 2021).

Evidence

(Jun et al., 2020)

Crush Optimism with Pessimism: Structured Bandits Beyond Asymptotic Optimality

> *"Structured bandits, generally defined, consist of bandit problems where there exist pieces of sideinformation or constraints in (and among) the mean rewards. While, of course, the standard K-armed bandit problem (Thompson, 1933) is a special case, researchers usually use the term structured bandits for nontrivial structure"*
>
> *"Many works in the structured bandits focus on specific parametric reward models, such as linear and generalized linear models (Rusmevichientong et al., 2008)(Abbasi-Yadkori et al., 2011)14,17]"*
>
> *"Russo and Van Roy (Russo et al., 2013) propose a general notion called eluder dimension that facilitates analysis for structured bandits for general function classes. For nonlinear reward models, worst-case regret guarantees have been obtained on reward structures such as unimodality (Yu et al., 2011), convexity [2], and Lipschitzness (Kleinberg, 2004)"*
>
> *".In the finite-sample regime, instance-optimal algorithms under specific model classes have been developed, such as unimodal reward [12] and Lipschitz rewards [30] have been developed"*
>
> *".Many linear bandit studies obtain regret guarantees that depend on the gap Δ between the best mean reward and the second-best mean reward [14,17,(Abbasi-Yadkori et al., 2011). As another example, in Lipschitz bandits, regret guarantees that depend on the zooming dimension or near-optimality dimension have been shown [24]9]."*

(Gupta et al., 2019)

Multi-Armed Bandits With Correlated Arms

> *"Structured bandits is a class of problems that cover linear bandits (Abbasi-Yadkori et al., 2011), generalized linear bandits (Filippi et al., 2010), Lipschitz bandits [20], global bandits (Atan et al., 2014), regional bandits (Wang et al., 2018) etc. In the structured bandits setup, mean rewards corresponding to different arms are related to one another through a hidden parameter θ. The underlying value of θ is fixed and the mean reward mappings θ μ k (θ) are known. Similarly, (Pandey et al., 2007) studies a dependent armed bandit problem, that also has mean rewards corresponding to different arms related to one another"*
>
> *".1. In this work we explicitly model the correlations in the rewards of a user corresponding to different arms. While mean rewards are related to each other in structured bandits and (Pandey et al., 2007), the reward realizations are not necessarily correlated"*
>
> *".2. The model studied here is non-parametric in the sense that there is no hidden feature space as is the case in structured bandits and the work of Pandey et al. (Pandey et al., 2007)"*
>
> *".4. Similar to the structured bandits, the unimodal bandit framework (Combes et al., 2014)(Trinh et al., 2019) also assumes a structure on the mean rewards and does not capture the reward correlations explicitly."*

(Gupta et al., 2021)

Best-Arm Identification in Correlated Multi-Armed Bandits

> *"A rare exception is the structured bandit setting, where mean rewards corresponding to different arms are related to one another through a hidden parameter θ. The underlying value of θ is fixed and unknown, but the mean reward mappings θ μ k (θ) are known. The linear bandit setting is a special case of structured bandits, where mean reward mappings are of the form x k θ with x k known to the player. The best-arm identification problem has been studied in (Soare et al., 2014)(Tao et al., 2018) for linear bandits and in (Huang et al., 2017) for the general structured bandit setting. Other special cases of structured bandits include global bandits (Atan et al., 2014), regional bandits (Wang et al., 2018) and the generalized linear bandits (Shen et al., 2018); to the best of our knowledge the best arm identification problem has not been addressed in these special cases. Note that in the full generality, the structured bandit framework is simply a bandit problem with constraints on the joint probability distribution [36], but that setting has only been studied for the objective of regret minimization and not best-arm identification. To the best of our knowledge, the structured bandits work studying best-arm identification (Soare et al., 2014)(Tao et al., 2018)(Huang et al., 2017) assume the presence of a hidden parameter θ through which mean rewards of different arms are related to one another."*

(Zhang, 2024)

Real-world Applications of Bandit Algorithms: Insights and Innovations

"In the problem of structured bandits, there exists a clear or implicit structure governing how rewards are allocated across different actions. Unlike traditional Multi-Armed Bandit (MAB) problems that focus on identifying the best arm to maximize rewards, structured bandit problems delve deeper into understanding the underlying reward structure. This understanding facilitates more informed decision-making, not just to maximize immediate rewards but also to unravel the reward mechanism itself, thereby optimizing long-term performance. Linear Structured Bandits assume that the reward for each action is a linear function of hidden parameters. The core challenge in this process is to in accurately estimate these parameters to predict the rewards for different actions. Early decision-making may involve selecting actions that are highly informative about the parameters, even if they don't offer the highest immediate reward. In Linear Structured Bandits problems, even if it initially selects sub-optimal arms, better long-term optimization is still possible as the algorithm becomes confident about the parameter. Global Structured Bandits deal with a more complex scenario where the reward structure can be nonlinear and intricate and task is still to understand the structure to make informed decisions. Success in global structured bandits depends on deciphering this structure to enhance decision-making and optimize rewards."

(Xu et al., 2018)

Multi-Armed Bandits on Partially Revealed Unit Interval Graphs

"Related work includes Lipschitz bandits (Agrawal, 1995), (Kleinberg et al., 2008), (Magureanu et al., 2014), taxonomy bandits (Slivkins, 2011) and unimodal bandits (Combes et al., 2014). Specifically, in Lipschitz bandits, the mean reward is assumed to be a Lipschitz function of the arm parameter"

"Recently, a general formulation of structured bandits was proposed in [23], which includes a variety of known bandit models (e.g., Lipschitz bandits, unimodal bandits, linear bandits, etc.) as well as the bandit model studied in this work as special cases."

(Kleinberg et al., 2008)

Multi-armed bandits in metric spaces

"In a multi-armed bandit problem, an online algorithm chooses from a set of strategies in a sequence of n trials so as to maximize the total payoff of the chosen strategies. While the performance of bandit algorithms with a small finite strategy set is quite well understood, bandit problems with large strategy sets are still a topic of very active investigation, motivated by practical applications such as online auctions and web advertisement. The goal of such research is to identify broad and natural classes of strategy sets and payoff functions which enable the design of efficient solutions. In this work we study a very general setting for the multi-armed bandit problem in which the strategies form a metric space, and the payoff function satisfies a Lipschitz condition with respect to the metric. We refer to this problem as the "Lipschitz MAB problem". We present a complete solution for the multi-armed problem in this setting. That is, for every metric space (L,X) we define an isometry invariant Max Min COV(X) which bounds from below the performance of Lipschitz MAB algorithms for X, and we present an algorithm which comes arbitrarily close to meeting this bound. Furthermore, our technique gives even better results for benign payoff functions."

(Combes et al., 2014)

Unimodal Bandits: Regret Lower Bounds and Optimal Algorithms

"We consider stochastic multi-armed bandits where the expected reward is a unimodal function over partially ordered arms. This important class of problems has been recently investigated in (Cope, 2009; Yu & Mannor, 2011). The set of arms is either discrete, in which case arms correspond to the vertices of a finite graph whose structure represents similarity in rewards, or continuous, in which case arms belong to a bounded interval. For discrete unimodal bandits, we derive asymptotic lower bounds for the regret achieved under any algorithm, and propose OSUB, an algorithm whose regret matches this lower bound. Our algorithm optimally exploits the unimodal structure of the problem, and surprisingly, its asymptotic regret does not depend on the number of arms. We also provide a regret upper bound for OSUB in nonstationary environments where the expected rewards smoothly evolve over time. The analytical results are supported by numerical experiments showing that OSUB performs significantly better than the state-of-the-art algorithms. For continuous sets of arms, we provide a brief discussion. We show that combining an appropriate discretization of the set of arms with the UCB algorithm yields an order-optimal regret, and in practice, outperforms recently proposed algorithms designed to exploit the unimodal structure."

(Rusmevichientong et al., 2008)

Linearly Parameterized Bandits

"We consider bandit problems involving a large (possibly infinite) collection of arms, in which the expected reward of each arm is a linear function of an r-dimensional random vector Z ∈ Rr, where r ≥ 2. The objective is to minimize the cumulative regret and Bayes risk. When the set of arms corresponds to the unit sphere, we prove that the regret and Bayes risk is of order Θ(r √T), by establishing a lower bound for an arbitrary policy, and showing that a matching upper bound is obtained through a policy that alternates between exploration and exploitation phases. The phase-based policy is also shown to be effective if the set of arms satisfies a strong convexity condition. For the case of a general set of arms, we describe a near-optimal policy whose regret and Bayes risk admit upper bounds of the form O(r √T log3/2T)."

(Atan et al., 2014)

Global Multi-armed Bandits with Hölder Continuity

> *"Standard Multi-Armed Bandit (MAB) problems assume that the arms are independent. However, in many application scenarios, the information obtained by playing an arm provides information about the remainder of the arms. Hence, in such applications, this informativeness can and should be exploited to enable faster convergence to the optimal solution. In this paper, formalize a new class of multi-armed bandit methods, Global Multi-armed Bandit (GMAB), in which arms are globally informative through a global parameter, i.e., choosing an arm reveals information about all the arms. We propose a greedy policy for the GMAB which always selects the arm with the highest estimated expected reward, and prove that it achieves bounded parameter-dependent regret. Hence, this policy selects suboptimal arms only nitely many times, and after a nite number of initial time steps, the optimal arm is selected in all of the remaining time steps with probability one. In addition, we also study how the informativeness of the arms about each other's rewards aects the speed of learning. Specically, we prove that the parameter-free (worst-case) regret is sublinear in time, and decreases with the informativeness of the arms. We also prove a sublinear in time Bayesian risk bound for the GMAB which reduces to the well-known Bayesian risk bound for linearly parameterized bandits when the arms are fully informative. GMABs have applications ranging from drug dosage control to dynamic pricing."*

(Wang et al., 2018)

Regional Multi-Armed Bandits

> *"We consider a variant of the classic multi-armed bandit problem where the expected reward of each arm is a function of an unknown parameter. The arms are divided into different groups, each of which has a common parameter. Therefore, when the player selects an arm at each time slot, information of other arms in the same group is also revealed. This regional bandit model naturally bridges the non-informative bandit setting where the player can only learn the chosen arm, and the global bandit model where sampling one arms reveals information of all arms. We propose an efficient algorithm, UCB-g, that solves the regional bandit problem by combining the Upper Confidence Bound (UCB) and greedy principles. Both parameter-dependent and parameter-free regret upper bounds are derived. We also establish a matching lower bound, which proves the order-optimality of UCB-g. Moreover, we propose SW-UCB-g, which is an extension of UCB-g for a non-stationary environment where the parameters slowly vary over time."*

(Soare et al., 2014)

Best-Arm Identification in Linear Bandits

> *"We study the best-arm identification problem in linear bandit, where the rewards of the arms depend linearly on an unknown parameter θ* and the objective is to return the arm with the largest reward. We characterize the complexity of the problem and introduce sample allocation strategies that pull arms to identify the best arm with a fixed confidence, while minimizing the sample budget. In particular, we show the importance of exploiting the global linear structure to improve the estimate of the reward of near-optimal arms. We analyze the proposed strategies and compare their empirical performance. Finally, as a by-product of our analysis, we point out the connection to the G-optimality criterion used in optimal experimental design."*

(Huang et al., 2017)

Structured Best Arm Identification with Fixed Confidence

> *"We study the problem of identifying the best action among a set of possible options when the value of each action is given by a mapping from a number of noisy micro-observables in the so-called fixed confidence setting. Our main motivation is the application to the minimax game search, which has been a major topic of interest in artificial intelligence. In this paper we introduce an abstract setting to clearly describe the essential properties of the problem. While previous work only considered a two-move game tree search problem, our abstract setting can be applied to the general minimax games where the depth can be non-uniform and arbitrary, and transpositions are allowed. We introduce a new algorithm (LUCB-micro) for the abstract setting, and give its lower and upper sample complexity results. Our bounds recover some previous results, which were only available in more limited settings, while they also shed further light on how the structure of minimax problems influence sample complexity."*

Linear and other parametric structured bandits

Linear and other parametric bandits assume rewards come from a small shared model, often a linear function of action or context features. This gives much stronger cross-arm generalization than Lipschitz methods: one pull can update estimates for many or even all actions at once, which can greatly reduce regret when the model is right. (11 sources)

The main parametric alternative to metric smoothness is to assume that expected rewards are generated by a **shared low-dimensional model**. In contextual settings, this often takes the linear form $\mathbb{E} = \langle \phi(x_t,a), \theta^\star\rangle$, so the learner does not estimate one mean per arm, but one unknown parameter vector that controls reward across all actions and contexts. This is why linear contextual bandits are usually treated as one of the three major structured families, alongside Lipschitz and policy-class approaches. (Banihashem et al., 2023) (Jiang et al., 2024)

This assumption gives **global information sharing**. A sample from one action updates the estimate of \theta^\star, and that update changes the predicted reward of every action whose features depend on the same parameter. Compared with a tabular learner, which must learn each arm almost separately, a linear learner can generalize immediately across a very large or even infinite action set as long as actions are represented by informative features. That is why Lipschitz, linear, and Gaussian-process bandit policies are commonly grouped together as methods for large-arm problems under reward-function constraints, in contrast to small-arm tabular methods. (Everett et al., 2025)

The regret gains can be dramatic when the parametric model is correct. Jiang et al. summarize a broad line of work establishing learning guarantees for **linear contextual bandits** in both low- and high-dimensional regimes. (Jiang et al., 2024) Important examples include early finite-time analyses for linear contextual bandits, minimax-optimal logarithmic regret for two-armed linear response models with forced sampling, and high-dimensional methods that use sparsity to stay efficient when many observed features are irrelevant. (Jiang et al., 2024) (Goldenshluger et al., 2013) (Bastani et al., 2020) (Qian et al., 2022)

A notable extension is **high-dimensional linear bandits with sparse structure**. Here the ambient covariate vector can be large, but only a small subset of features matters. Bastani et al. show that a LASSO-based bandit algorithm can achieve near-oracle performance in this setting, and Qian et al. give multi-stage elimination and randomization methods with near-optimal finite-time regret that can also adapt to margin and gap conditions. This again highlights the central theme of structured learning: the learner exploits a compressed model of reward rather than paying separately for each nominal action-context pair. (Bastani et al., 2020) (Qian et al., 2022)

Parametric structure can also reduce the need for explicit exploration. In standard bandit thinking, greedy behavior is often unsafe, but Bastani et al. show that in linear contextual problems with enough randomness in the observed covariates, a simple greedy algorithm can already be rate-optimal, and their Greedy-First method explores only when the data do not provide enough natural coverage. So, unlike tabular bandits where direct exploration of each arm is usually unavoidable, parametric models can sometimes get much of their learning “for free” from variation in the contexts. (Bastani et al., 2017)

The contrast with nonparametric models is helpful. Jiang et al. place linear contextual bandits on one side and smooth nonparametric bandits on the other: linear models make a stronger functional assumption and can therefore get faster, more global transfer, while nonparametric methods use weaker smoothness assumptions and pay rates that depend more heavily on dimension or local complexity. (Jiang et al., 2024) Hu et al. make this bridge explicit by viewing smooth contextual bandits as lying between two extremes: **nondifferentiable/local** problems where one must learn mostly region by region, and **parametric-response** problems where there is effectively infinite extrapolatability across contexts. (Hu et al., 2019)

This same local-versus-global tradeoff appears in contextual nonparametric methods. Algorithms such as ABSE and k-NN UCB generalize across nearby contexts using smoothness or metric assumptions, and can adapt to intrinsic dimension or margin conditions, but they still learn in a much more local way than linear bandits do. In a linear model, two far-apart actions or contexts may still strongly inform each other if they share the same parameter; in a nonparametric smooth model, transfer usually weakens with distance. (Perchet et al., 2011) (Reeve et al., 2018) (Guan et al., 2018)

Overall, the key lesson is that **parametric structured bandits replace “similar actions have similar rewards” with “all rewards obey a common formula.”** When that formula is well chosen—linear, sparse linear, generalized linear, or another low-dimensional parametric family—the learner can generalize much more aggressively than either tabular or purely local metric methods, often leading to much faster learning in large action spaces. (Banihashem et al., 2023) (Jiang et al., 2024)

Evidence

(Banihashem et al., 2023)

An Improved Relaxation for Oracle-Efficient Adversarial Contextual Bandits

"There are three prominent problem setups broadly studied in the contextual bandits literature: Lipschitz contextual bandits, linear contextual bandits, and contextual bandits with policy class. Lipschitz contextual bandits (Pál et al., 2008)(Cutkosky et al., 2017) and linear contextual bandits (Chu et al., 2011) assume a structured payoff based on a Lipschitz or linear realizability assumption, respectively. The strong structural assumptions made by these works however make them impractical for many settings."

(Jiang et al., 2024)

Batched Nonparametric Contextual Bandits

"On one hand, (Auer, 2003)(Abbasi-Yadkori et al., 2011)(Goldenshluger et al., 2013)(Bastani et al., 2020)(Bastani et al., 2017)(Qian et al., 2022) obtained learning guarantees for linear contextual bandits in both low and high dimensional settings. On the other hand, (Yang et al., 2002) introduced the nonparametric approach to model the mean reward function. [50] proved a minimax lower bound on the regret of nonparametric bandit and developed an upper-confidence-bound (UCB) based policy to achieve a near-optimal rate. (Perchet et al., 2011) improved this result and proposed the Adaptively Binned Successive Elimination (ABSE) policy that can also adapt to the unknown margin parameter. Further insights in this nonparametric setting were developed in subsequent works (Qian et al., 2016)(Qian et al., 2016)(Reeve et al., 2018)(Guan et al., 2018)(Hu et al., 2019)(Suk et al., 2020)25,10,54,9]. The smoothness assumption is also adopted in another line of work (Pál et al., 2008)(Locatelli et al., 2018)(Krishnamurthy et al., 2019)(Cai et al., 2022) on the continuum-armed bandit problems. However in contrast to what we study, the reward is assumed to be a Lipschitz function of the action, and the covariates are not taken into considerations"

".(Feng et al., 2021) considered Lipschitz continuum-armed bandit problem with the batch constraint."

(Everett et al., 2025)

Quick Draw Bandits: Quickly Optimizing in Nonstationary Environments with Extremely Many Arms

"existing non-stationary bandit policies are designed for a small number of arms, while Lipschitz, linear, and Gaussian process bandit policies are designed to handle a large (or infinite) number of arms in stationary reward environments under constraints on the reward function."

(Goldenshluger et al., 2013)

A Linear Response Bandit Problem

"We consider a two–armed bandit problem which involves sequential sampling from two non-homogeneous populations. The response in each is determined by a random covariate vector and a vector of parameters whose values are not known a priori. The goal is to maximize cumulative expected reward. We study this problem in a minimax setting, and develop rate-optimal polices that combine myopic action based on least squares estimates with a suitable "forced sampling" strategy. It is shown that the regret grows logarithmically in the time horizon n and no policy can achieve a slower growth rate over all feasible problem instances. In this setting of linear response bandits, the identity of the sub-optimal action changes with the values of the covariate vector, and the optimal policy is subject to sampling from the inferior population at a rate that grows like n."

(Bastani et al., 2020)

Online Decision Making with High-Dimensional Covariates

"Big data has enabled decision-makers to tailor decisions at the individual-level in a variety of domains such as personalized medicine and online advertising. This involves learning a model of decision rewards conditional on individual-specific covariates. In many practical settings, these covariates are high-dimensional; however, typically only a small subset of the observed features are predictive of a decision's success. We formulate this problem as a multi-armed bandit with high-dimensional covariates, and present a new efficient bandit algorithm based on the LASSO estimator. Our regret analysis establishes that our algorithm achieves near-optimal performance in comparison to an oracle that knows all the problem parameters. The key step in our analysis is proving a new oracle inequality that guarantees the convergence of the LASSO estimator despite the non-i.i.d. data induced by the bandit policy. Furthermore, we illustrate the practical relevance of our algorithm by evaluating it on a real-world clinical problem of warfarin dosing. A patient's optimal warfarin dosage depends on the patient's genetic profile and medical records; incorrect initial dosage may result in adverse consequences such as stroke or bleeding. We show that our algorithm outperforms existing bandit methods as well as physicians to correctly dose a majority of patients."

(Qian et al., 2022)

Adaptive Algorithm for Multi-Armed Bandit Problem with High-Dimensional Covariates

"Abstract This article studies an important sequential decision making problem known as the multi-armed stochastic bandit problem with covariates. Under a linear bandit framework with high-dimensional covariates, we propose a general multi-stage arm allocation algorithm that integrates both arm elimination and randomized assignment strategies. By employing a class of high-dimensional regression methods for coefficient estimation, the proposed algorithm is shown to have near optimal finite-time regret performance under a new study scope that requires neither a margin condition nor a reward gap condition for competitive arms.

Based on the synergistically verified benefit of the margin, our algorithm exhibits adaptive performance that automatically adapts to the margin and gap conditions, and attains optimal regret rates simultaneously for both study scopes, without or with the margin, up to a logarithmic factor. Besides the desirable regret performance, the proposed algorithm simultaneously generates useful coefficient estimation output for competitive arms and is shown to achieve both estimation consistency and variable selection consistency. Promising empirical performance is demonstrated through extensive simulation and two real data evaluation examples. Supplementary materials for this article are available online."

(Bastani et al., 2017)

Mostly Exploration-Free Algorithms for Contextual Bandits

"The contextual bandit literature has traditionally focused on algorithms that address the exploration–exploitation tradeoff. In particular, greedy algorithms that exploit current estimates without any exploration may be suboptimal in general. However, exploration-free greedy algorithms are desirable in practical settings where exploration may be costly or unethical (e.g., clinical trials). Surprisingly, we find that a simple greedy algorithm can be rate optimal (achieves asymptotically optimal regret) if there is sufficient randomness in the observed contexts (covariates). We prove that this is always the case for a two-armed bandit under a general class of context distributions that satisfy a condition we term covariate diversity. Furthermore, even absent this condition, we show that a greedy algorithm can be rate optimal with positive probability. Thus, standard bandit algorithms may unnecessarily explore. Motivated by these results, we introduce Greedy-First, a new algorithm that uses only observed contexts and rewards to determine whether to follow a greedy algorithm or to explore. We prove that this algorithm is rate optimal without any additional assumptions on the context distribution or the number of arms. Extensive simulations demonstrate that Greedy-First successfully reduces exploration and outperforms existing (exploration-based) contextual bandit algorithms such as Thompson sampling or upper confidence bound. This paper was accepted by J. George Shanthikumar, big data analytics."

(Hu et al., 2019)

Smooth Contextual Bandits: Bridging the Parametric and Non-differentiable Regret Regimes

"Dynamic Personalized Decision Making Beyond the Super-Extrapolatable and Super-Local Cases Contextual bandit problems model the inherent trade-off between exploration and exploitation in personalized decision making in marketing, healthcare, revenue management, and more. Specifically, the trade-off is characterized by the optimal growth rate of the regret. Intuitively, the optimal rate should depend on how complex the underlying supervised learning problem is, namely, how much can observing reward in one context tell us about mean rewards in another. To formalize this intuitive relationship, Hu, Kallus, and Mao study in "Smooth Contextual Bandits: Bridging the Parametric and Nondifferentiable Regimes" a nonparametric contextual bandit problem in which mean reward functions are β-times differentiable (more generally, Hölder β-smooth). This interpolates between two extremes previously studied in isolation: nondifferentiable bandits (β ≤ 1), with which running separated noncontextual bandits in different context regions achieves rate-optimal regret, and parametric-response bandits (β = ∞), with which rate-optimal regret can be achieved with minimal or no exploration because of infinite extrapolatability across contexts. The authors develop a rate-optimal algorithm that operates neither fully locally nor fully globally, revealing the optimal regret rate in this in-between smooth setting and shedding light on the crucial interplay of functional complexity and regret in dynamic personalized decision making."

(Perchet et al., 2011)

The multi-armed bandit problem with covariates

"We consider a multi-armed bandit problem in a setting where each arm produces a noisy reward realization which depends on an observable random covariate. As opposed to the traditional static multi-armed bandit problem, this setting allows for dynamically changing rewards that better describe applications where side information is available. We adopt a nonparametric model where the expected rewards are smooth functions of the covariate and where the hardness of the problem is captured by a margin parameter. To maximize the expected cumulative reward, we introduce a policy called Adaptively Binned Successive Elimination (abse) that adaptively decomposes the global problem into suitably "localized" static bandit problems. This policy constructs an adaptive partition using a variant of the Successive Elimination (se) policy. Our results include sharper regret bounds for the se policy in a static bandit problem and minimax optimal regret bounds for the abse policy in the dynamic problem."

(Reeve et al., 2018)

The K-Nearest Neighbour UCB Algorithm for Multi-Armed Bandits with Covariates

"In this paper we propose and explore the k-Nearest Neighbour UCB algorithm for multi-armed bandits with covariates. We focus on a setting where the covariates are supported on a metric space of low intrinsic dimension, such as a manifold embedded within a high dimensional ambient feature space. The algorithm is conceptually simple and straightforward to implement. The k-Nearest Neighbour UCB algorithm does not require prior knowledge of the either the intrinsic dimension of the marginal distribution or the time horizon. We prove a regret bound for the k-Nearest Neighbour UCB algorithm which is minimax optimal up to logarithmic factors. In particular, the algorithm automatically takes advantage of both low intrinsic dimensionality of the marginal distribution over the covariates and low noise in the data, expressed as a margin condition. In addition, focusing on the case of bounded rewards, we give corresponding regret bounds for the k-Nearest Neighbour KL-UCB algorithm, which is an analogue of the KL-UCB algorithm adapted to the setting of multi-armed bandits with covariates. Finally, we present empirical results which demonstrate the ability of both the k-Nearest Neighbour UCB and k-Nearest Neighbour KL-UCB to take advantage of situations where the data is

supported on an unknown sub-manifold of a high-dimensional feature space."

(Guan et al., 2018)

Nonparametric Stochastic Contextual Bandits

"We analyze the K-armed bandit problem where the reward for each arm is a noisy realization based on an observed context under mild nonparametric assumptions.We attain tight results for top-arm identification and a sublinear regret of Õ(T1+D/(2+D), where D is the context dimension, for a modified UCB algorithm that is simple to implement. We then give global intrinsic dimension dependent and ambient dimension independent regret bounds. We also discuss recovering topological structures within the context space based on expected bandit performance and provide an extension to infinite-armed contextual bandits. Finally, we experimentally show the improvement of our algorithm over existing approaches for both simulated tasks and MNIST image classification."

Gaussian process, kernel, and other nonlinear correlated-action models

Gaussian process and kernel bandits are nonlinear structured bandits that model rewards as correlated across actions, not independent. They let one observation update beliefs about many other actions at once, often much more flexibly than linear models, but they rely on stronger prior assumptions about smoothness or function class. (6 sources)

A key step beyond linear bandits is to model reward as an **unknown nonlinear function with correlation across actions**. Tsai et al. place **GP-UCB** and **KernelUCB** in exactly this role: GP-UCB assumes the reward function is drawn from a Gaussian process with known mean and covariance kernel, while KernelUCB assumes the reward function lies in a reproducing kernel Hilbert space (RKHS) with bounded norm. They present these methods as ways to handle nonlinearity when linear or generalized linear models are too restrictive, while also noting that they require fairly strong assumptions on the reward function. (Tsai et al., 2022)

The main structural idea is that the learner uses a **kernel or GP prior to encode similarity**, so rewards at related actions are statistically tied together. This is a different form of sharing than in Lipschitz bandits: instead of only saying nearby actions must have similar rewards in a metric sense, the kernel gives a richer covariance pattern that can capture smoothness, periodicity, or other nonlinear relations. In this sense, GP and kernel bandits sit between local metric methods and global parametric models: they still generalize across the whole action space, but the strength of that generalization is controlled by the kernel rather than by a finite-dimensional parameter vector. (Tsai et al., 2022) (Zhou et al., 2025)

This gives a direct answer to the comparison with tabular learners. In a tabular bandit, pulling one arm mainly teaches the learner about that arm alone. In GP or kernel bandits, a pull at one action changes the posterior mean and uncertainty of many other actions, especially those the kernel views as similar, so it can be wasteful to maintain a large nominal action set as if every action were unrelated. Zhou et al. make this motivation explicit: in many natural large-action problems, actions are highly correlated and many have similar rewards across environments, so treating them independently wastes information. (Zhou et al., 2025)

The notable algorithmic example is **GP-UCB**, which applies optimism under uncertainty using GP posterior mean and variance. As summarized by Zhou et al., Srinivas et al. model the reward function as a Gaussian process prior and prove a cumulative-regret bound of order O(T \gamma_T \log |A_{\mathrm{full}}|) for finite action spaces, where \gamma_T is the maximum information gain after T rounds; later work extended GP-style guarantees to continuous domains. (Zhou et al., 2025) (Chowdhury et al., 2017) A useful feature of these bounds is that they depend on **information gain** under the kernel, not directly on the number of possible actions, which is one reason these methods are attractive for very large or continuous spaces. (Zhou et al., 2025) (Chowdhury et al., 2017)

Later GP work sharpened this picture. Chowdhury and Gopalan give **IGP-UCB** and **GP-TS** for continuous-arm optimization, with regret guarantees when the true reward function belongs to the RKHS associated with the chosen kernel. This is important because it shows that GP-style exploration can be analyzed without assuming a literal Bayesian data-generating prior; it is enough that the function is well behaved in the kernel class. (Chowdhury et al., 2017) More recent batched versions also get near-optimal regret of order O^*(\sqrt{T\gamma_T}) under RKHS assumptions, again showing that kernelized correlation can make large

continuous action spaces manageable. (Li et al., 2021)

These models are best seen as one major branch of **nonlinear structured bandits**. Tsai et al. distinguish them from **generalized linear bandits**, where the reward is a link function applied to a linear predictor, and from more geometry-based methods like HOO and zooming. So the landscape is now fairly clear: generalized linear models give a specific parametric nonlinearity; Lipschitz and HOO-style methods use local smoothness in a metric or dissimilarity space; GP and kernel methods use covariance structure to correlate actions in a more flexible way. (Tsai et al., 2022) (Bubeck et al., 2010) (Kleinberg et al., 2008)

The tradeoff is the usual one in structured learning. Compared with tabular methods, GP and kernel bandits can generalize far more aggressively across similar actions. Compared with linear models, they can fit much richer nonlinear reward patterns. But this flexibility comes only if the chosen kernel or RKHS really matches the reward structure well enough, which is why the literature repeatedly notes that these methods rely on fairly strong modeling assumptions. (Tsai et al., 2022)

Evidence

(Tsai et al., 2022)

Fast online inference for nonlinear contextual bandit based on Generative Adversarial Network

> *"Alongside there are many existing structures investigated, including: linear, combinatorial, Lipschitz, and unimodal bandits [8]. There are also settings with infinitely many arms (Abbasi-Yadkori et al., 2011)(Wang et al., 2008) and in generic continuous metric space [16,(Bubeck et al., 2010) with hierarchical tree-based space partition algorithms. To deal with nonlinearity, generalized linear bandits have been considered. GLM-UCB (Filippi et al., 2010) assumes that the reward function can be written as a composition of a linear function and a link function. Others explore more general nonlinear bandits without making strong modeling assumptions. GP-UCB [10] assumes that the reward function is generated from a Gaussian process with known mean and covariance functions. KernelUCB [18] assumes that the reward function lies in a RKHS with bounded RKHS norm. Nevertheless, these methods require fairly strong assumptions on the reward function."*

(Zhou et al., 2025)

Representative Action Selection for Large Action Space Bandit Families

> *"Lipschitz Bandits study bandit problems over continuous or very large action spaces, under the assumption that the reward function varies smoothly with respect to a metric on the action space (Kleinberg et al., 2008)(Slivkins, 2009). Common approaches include the Zooming algorithm (Kleinberg et al., 2008), which refines the exploration process according to the local geometry of nearoptimal regions rather than relying on uniform discretization, achieving regret tied to the "zooming dimension"; and Hierarchical Optimistic Optimization [Bubeck et al., 2011], which maintains a binary tree whose nodes correspond to measurable regions of the action space, explores by traversing the tree according to optimistic values, and progressively expands nodes to refine the partition in near-optimal regions. This line of work is closely related to ours, as both deal with large action spaces endowed with a similarity structure, and the Zooming algorithm likewise exploits the notion of □ nets"*

> *". Srinivas et al. [2009] model the reward function as a sample from a Gaussian process prior and propose the GP-UCB algorithm, which achieves cumulative regret O T γ T log |A full | for finite action spaces, where γ T denotes the maximum information gain after T rounds. The widespread adoption of this approach-both for finite action sets [Valko et al., 2013, Li andScarlett, 2022] and continuous domains [Chowdhury and Gopalan, 2017] highlights the practicality of assuming that action outcomes are correlated"*

> *". In many natural situations, while the nominal set of actions is large, actions are highly correlated: many yield similar rewards across environments, making it wasteful to maintain the full set."*

(Chowdhury et al., 2017)

On Kernelized Multi-armed Bandits

> *"We consider the stochastic bandit problem with a continuous set of arms, with the expected reward function over the arms assumed to be fixed but unknown. We provide two new Gaussian process-based algorithms for continuous bandit optimization-Improved GP-UCB (IGP-UCB) and GP-Thomson sampling (GP-TS), and derive corresponding regret bounds. Specifically, the bounds hold when the expected reward function belongs to the reproducing kernel Hilbert space (RKHS) that naturally corresponds to a Gaussian process kernel used as input by the algorithms. Along the way, we derive a new self-normalized concentration inequality for vector- valued martingales of arbitrary, possibly infinite, dimension. Finally, experimental evaluation and comparisons to existing algorithms on synthetic and real-world environments are carried out that highlight the favorable gains of the proposed strategies in many cases."*

(Li et al., 2021)

Gaussian Process Bandit Optimization with Few Batches

> *"In this paper, we consider the problem of black-box optimization using Gaussian Process (GP) bandit optimization with a small number of batches. Assuming the unknown function has a low norm in the Reproducing Kernel Hilbert Space (RKHS), we introduce a batch algorithm inspired by batched finite-arm bandit algorithms, and show that it achieves the cumulative regret upper bound $O^\ast(\sqrt{T\gamma_T})$ using $O(\log\log T)$ batches within time horizon T, where the $O^\ast(\cdot)$ notation hides dimension-independent logarithmic factors and γ_T is the maximum information gain associated with the kernel. This bound is near-optimal for several kernels of interest and improves on the typical $O^\ast(\sqrt{T}\gamma_T)$ bound, and our approach is arguably the simplest among algorithms attaining this improvement. In addition, in the case of a constant number of batches (not depending on T), we propose a modified version of our algorithm, and characterize how the regret is impacted by the number of batches, focusing on the squared exponential and Mat\'ern kernels. The algorithmic upper bounds are shown to be nearly minimax optimal via analogous algorithm-independent lower bounds."*

(Bubeck et al., 2010)

X-Armed Bandits

> *"We consider a generalization of stochastic bandits where the set of arms, X, is allowed to be a generic measurable space and the mean-payoff function is "locally Lipschitz" with respect to a dissimilarity function that is known to the decision maker. Under this condition we construct an arm selection policy, called HOO (hierarchical optimistic optimization), with improved regret bounds compared to previous results for a large class of problems. In particular, our results imply that if X is the unit hypercube in a Euclidean space and the mean-payoff function has a finite number of global maxima around which the behavior of the function is locally continuous with a known smoothness degree, then the expected regret of HOO is bounded up to a logarithmic factor by √n, that is, the rate of growth of the regret is independent of the dimension of the space. We also prove the minimax optimality of our algorithm when the dissimilarity is a metric. Our basic strategy has quadratic computational complexity as a function of the number of time steps and does not rely on the doubling trick. We also introduce a modified strategy, which relies on the doubling trick but runs in linearithmic time. Both results are improvements with respect to previous approaches."*

(Kleinberg et al., 2008)

Multi-armed bandits in metric spaces

> *"In a multi-armed bandit problem, an online algorithm chooses from a set of strategies in a sequence of n trials so as to maximize the total payoff of the chosen strategies. While the performance of bandit algorithms with a small finite strategy set is quite well understood, bandit problems with large strategy sets are still a topic of very active investigation, motivated by practical applications such as online auctions and web advertisement. The goal of such research is to identify broad and natural classes of strategy sets and payoff functions which enable the design of efficient solutions. In this work we study a very general setting for the multi-armed bandit problem in which the strategies form a metric space, and the payoff function satisfies a Lipschitz condition with respect to the metric. We refer to this problem as the "Lipschitz MAB problem". We present a complete solution for the multi-armed problem in this setting. That is, for every metric space (L,X) we define an isometry invariant Max Min COV(X) which bounds from below the performance of Lipschitz MAB algorithms for X, and we present an algorithm which comes arbitrarily close to meeting this bound. Furthermore, our technique gives even better results for benign payoff functions."*

Ordinal action spaces and structured generalization versus tabular learning

Ordinal action spaces are action sets with an order, such as ratings, prices, doses, or positions on a line. They matter because order gives structure: nearby actions can often be treated as related, so one observation can help estimate several actions instead of only the one that was chosen. (4 sources)

Ordinal action spaces are one concrete example of the larger move from tabular to structured bandits. Wan et al. list **ordinal action spaces** alongside Lipschitz, linear, and Gaussian-process bandits as structured models, and they state the main contrast clearly: in tabular learning each action is treated independently, while in structured learning information from one action can generalize to others that are “similar” or “close” under the chosen structure (Wan et al., 2022). For ordinal actions, that structure is usually the fact that actions come with a natural ranking or location on a line, so adjacent actions are often more related than distant ones (Wan et al., 2022) (Model-Generated).

This makes ordinal spaces a natural fit for **local generalization**. If actions are doses, prices, or scalar predictions, then trying one value can say something about nearby values, much like in one-dimensional Lipschitz or continuum-armed problems. Saha et al. make this contrast explicit in a neighboring literature:

standard contextual-bandit methods are usually built for **discrete** action and policy spaces, whereas their pseudo-1d bandit setting handles a **continuous prediction/action space and infinite policy space** under convex Lipschitz assumptions (Saha et al., 2021). Likewise, Ju et al. note that continuous-armed bandits allow infinitely many actions, and that another branch assumes a **Lipschitz condition on expected rewards**, with adaptive zooming methods used to exploit this similarity structure rather than treating actions as unrelated (Ju et al., 2022) (Slivkins, 2009).

So the practical lesson is straightforward. When actions have an order or metric relation, a structured learner can use that relation to share evidence across neighboring or otherwise linked actions; a tabular learner cannot, because it keeps separate estimates for each action. This is why structured methods are preferred in large, continuous, or ordered action spaces: they can often learn from the **shape** of the action space instead of paying the full cost of learning each option one by one (Wan et al., 2022) (Slivkins, 2009).

Evidence

(Wan et al., 2022)

Towards Scalable and Robust Structured Bandits: A Meta-Learning Framework

> *"Lipschitz bandits and continuum-armed bandits are proposed for the metric and compact action spaces, respectively (Agrawal, 1995; Kleinberg et al., 2008). More generally, in structured bandits, one can formulate the model according to a broad variety of structures, such as linear bandits (Rusmevichientong and Tsitsiklis, 2010; Dani et al., 2008), Gaussian process bandits (Srinivas et al., 2010), and ordinal action spaces (Yue and Guestrin, 2011). In contrast to the tabular learning where each action is treated independently, the structured learner can generalize information from one action to the others as long as they are "similar" or "close" in terms of the pre-specified metric or structure."*

(Saha et al., 2021)

Optimal Regret Algorithm for Pseudo-1d Bandit Convex Optimization

> *"CB formulations work with general loss/reward functions while we restrict our methods to convex Lipschitz functions only. On the other hand, CB methods are designed in general for discrete action and policy space (see Remark 3 in Section 2) unlike pseudo-1d bandit formulation that handles continuous prediction/action space and infinite policy space."*

(Ju et al., 2022)

Dynamic path learning in decision trees using contextual bandits

> *"MAB algorithms are very versatile and can be adapted into a variety of scenarios. For example, some studies assume a linear relationship between the actions and the reward [14](Russac et al., 2019)(Ito et al., 2020), which is not applicable in our study; some studies investigate rotting bandit, where the expected reward of each arm decays as a function of the times it has been selected (Levine et al., 2017)[18]; sleeping bandit (Li et al., 2019)[20] investigates the scenario where the actions may not be available at all times; continuous-armed bandit [21](Hu et al., 2019)[23] allows the application to have an infinity amount of actions; cascading bandit (Kveton et al., 2015)(Li et al., 2017) allows the user to select the first attractive item out of a list of items"*

> *".Another branch of the study assumes Lipschitz condition on the expected rewards, which is adapted by our study. Standard Lipschitz contextual bandit is first introduced in (Hazan et al., 2007). (Slivkins, 2009)[36] further reduce the regret by adopting a zooming algorithm to adaptively partition the context space."*

(Slivkins, 2009)

Contextual Bandits with Similarity Information

> *"In a multi-armed bandit (MAB) problem, an online algorithm makes a sequence of choices. In each round it chooses from a time-invariant set of alternatives and receives the payoff associated with this alternative. While the case of small strategy sets is by now well-understood, a lot of recent work has focused on MAB problems with exponentially or infinitely large strategy sets, where one needs to assume extra structure in order to make the problem tractable. In particular, recent literature considered information on similarity between arms.*
>
> *We consider similarity information in the setting of contextual bandits, a natural extension of the basic MAB problem where before each round an algorithm is given the context--a hint about the payoffs in this round. Contextual bandits are directly motivated by placing advertisements on web pages, one of the crucial problems in sponsored search. A particularly simple way to represent similarity information in the contextual bandit setting is via a similarity distance between the context-arm pairs which bounds from above the difference between the respective expected payoffs.*

Prior work on contextual bandits with similarity uses "uniform" partitions of the similarity space, so that each context-arm pair is approximated by the closest pair in the partition. Algorithms based on "uniform" partitions disregard the structure of the payoffs and the context arrivals, which is potentially wasteful. We present algorithms that are based on adaptive partitions, and take advantage of "benign" payoffs and context arrivals without sacrificing the worst-case performance. The central idea is to maintain a finer partition in high-payoff regions of the similarity space and in popular regions of the context space. Our results apply to several other settings, e.g., MAB with constrained temporal change (Slivkins and Upfal, 2008) and sleeping bandits (Kleinberg et al., 2008a)."